

LECTURE 1 (SEPTEMBER 1ST)

Definition. (Density operator) A general quantum state is represented by a density operator.

$$\rho = \sum_{i=1}^n p_i |\psi_i\rangle \langle \psi_i|$$

where ρ is

- (i) Hermitian $\rho = \rho^\dagger$
- (ii) Positive operator, meaning that $\langle \psi | \rho | \psi \rangle \geq 0$ for an arbitrary $|\psi\rangle$
- (iii) Trace is 1, that is, $\text{Tr}(\rho) = 1$

It is easy to think the operator that gives classical probability weighting when a measurement is performed. For example, take an experiment that either has a spin up or spin down as its output. Generally, an output would be expressed as a superposition,

$$E[O] = \langle \psi_1 | \psi_1 \rangle p_1 \langle \psi_1 | O | \psi_1 \rangle + \langle \psi_2 | \psi_2 \rangle p_2 \langle \psi_2 | O | \psi_2 \rangle + \dots$$

Notice that by taking the above operator as the definition of a mixed state, we can simply take

$$\text{Tr}(\rho O)$$

to obtain the expected value of a certain observable. The thing is, there are two types of uncertainty – classical and quantum.

Theorem. (Condition for pure states and equivalences *) Notice that a pure state can be expressed as a density operator like the following

$$\hat{\rho} = |\psi\rangle \langle \psi|$$

These types of operators, called projection operators, are idempotent

$$\rho^2 = \rho$$

meaning that their expected value is either 1 or 0. Now notice that two density matrices of pure states are equivalent up to phase. We see that

$$|x\rangle \langle x| = |y\rangle \langle y|$$

if and only if

$$|x\rangle = |y\rangle \langle y | x \rangle$$

where

$$1 = |\langle x | y \rangle|^2$$

and the states differ purely by some phase factor $\exp(i\theta)$ as we would expect.

LECTURE 2 (SEPTEMBER 3RD)

Theorem. (Born rule in the language of density operators) Using the spectral decomposition of a certain state $A = \sum a|a\rangle\langle a| = \sum a \text{Pr}_a$, we can express the probability of observing A and obtaining a as

$$\boxed{\text{Tr}(\rho \text{Pr}_a)}$$

What would happen in the case of a pure state?

$$\frac{\text{Pr}_a \hat{\rho} \text{Pr}_a}{\text{Tr}(\hat{\rho} \text{Pr}_a)} = \frac{1}{\text{Tr}(\hat{\rho} \text{Pr}_a)} \text{Tr}(\text{Pr}_a \hat{\rho} \text{Pr}_a) = 1$$

and that the trace is 1.

Theorem. (Time evolution of the density operator) We check that the density matrix satisfies, for a pure state,

$$\begin{aligned} \frac{d\rho}{dt} &= \frac{d}{dt} (|\psi(t)\rangle\langle\psi(t)|) = \left(\frac{d}{dt} |\psi(t)\rangle \right) \langle\psi(t)| + |\psi(t)\rangle \left(\frac{d}{dt} \langle\psi(t)| \right) \\ &= -\frac{i}{\hbar} H |\psi\rangle\langle\psi| + \frac{i}{\hbar} |\psi\rangle\langle\psi| H = -\frac{i}{\hbar} [H, \rho] \end{aligned}$$

We that, therefore, the “objects that move” is the state.

Remark. (Classification of ρ) Notice that the square of the density matrix is equal to

$$\hat{\rho}^2 = \sum_j p_j^2 |\psi_j\rangle\langle\psi_j|$$

Therefore, in general, we say mixed states are states whose density matrix satisfies

$$\hat{\rho} \neq \hat{\rho}^2$$

and a pure state otherwise.

Proof. Let's compare ρ^2 and ρ

$$\rho - \rho^2 = \sum_j (p_j - p_j^2) |\psi_j\rangle\langle\psi_j|$$

and the expected value is always bigger than 0. Sometimes, we write $\rho \geq \rho^2$. We also ask, what is the norm of $|\rho\psi\rangle$ for an arbitrary $|\psi\rangle$? We notice that

$$\langle\rho\psi|\rho\psi\rangle = \langle\psi|\rho^2|\psi\rangle = \langle\psi|\rho\psi\rangle \leq \sqrt{\langle\psi|\psi\rangle} \sqrt{\langle\rho\psi|\rho\psi\rangle}$$

Resulting in

$$\langle \rho \psi | \rho \psi \rangle \leq \langle \psi | \psi \rangle$$

We see that the density matrix is a contraction mapping. We now prove the contraposition – if $\rho^2 = \rho$, the state is pure. Last class we have seen that the eigenvalue of the matrix becomes either 1 or 0. In the spectral decomposition,

$$\rho = \sum_j p_j |\psi_j\rangle \langle \psi_j|$$

we see that only one of the p_j 's should become one and that the others should be zero. We have now concluded the proof. $\square$

Theorem. (Alternative decompositions) Let's say that we have a certain mixed state. How do we find the relationship between a mixed state expressed in an arbitrary basis and a basis that is not necessarily orthogonal? Let say you have a orthogonal basis $\{|\psi_i\rangle\}$ where n is the number of eigenketes with non-zero eigenvalues. How does it map onto another representation?

$$\rho = \sum_{i=1}^n p_i |\psi_i\rangle \langle \psi_i| \quad \mapsto \quad \rho = \sum_{j=1}^m q_j |\varphi_j\rangle \langle \varphi_j|$$

We obviously know that $n \leq m$. Take

$$\langle \psi_k | \hat{\rho} | \psi_k \rangle = 0$$

Then,

$$\sum_{j=1}^N w_j |\langle \psi_i | \alpha_j \rangle|^2 = 0$$

which tells us that

$$\langle \psi_i | \alpha_j \rangle = 0$$

Therefore,

$$|\alpha_j\rangle = \sum_{k=1}^n \langle \psi_k | \alpha_j \rangle |\psi_k\rangle$$

We now define a $m \times n$ matrix

$$V_{ij} = \sqrt{\frac{w_i}{p_j}} \langle \psi_j | \alpha_i \rangle$$

It is obvious that

$$\sum_{j=1}^n V_{ij} \sqrt{p_j} |\psi_j\rangle = \sqrt{w_i} |\alpha_i\rangle$$

and we have

$$\sum_{i=1}^m (V^\dagger)_{ji} V_{ij} = \frac{1}{\sqrt{p_j p_k}} \sum_{i=1}^m w_i \langle \psi_j | \alpha_i \rangle \langle \alpha_i | \psi_k \rangle = \frac{1}{\sqrt{p_j p_k}} \langle \psi_j | \rho^2 | \psi_k \rangle = \delta_{jk}$$

LECTURE 3 (SEPTEMBER 8TH)

Definition. (Perturbation theory) In perturbation theory, an arbitrary Hamiltonian is obtained through writing

$$H(t) = H_0 + \lambda H_1(t)$$

where we fully know H_0 analytically. In general, we deal with

- (i) Time independent perturbations
- (ii) Time dependent perturbations

Example. (Non-degenerate perturbation) Given the information of H_0 we want to inquire the following,

$$E_n^{(0)} \quad \& \quad |n^{(0)}\rangle \quad \implies \quad |n\rangle_\lambda \quad \& \quad E_n$$

where $\hat{H}|n\rangle_\lambda = E_n|n\rangle_\lambda$. We are to of course expect the following relationship for an arbitrary function $\Delta_n(\lambda)$

$$E_n = E_n^{(0)} + \Delta_n(\lambda)$$

and thus

$$(\hat{H}_0 - E_n^{(0)})|n\rangle_\lambda = (\Delta_n(\lambda) - \lambda\hat{V})|n\rangle_\lambda$$

As $H_0|n^{(0)}\rangle = E_n^{(0)}|n^{(0)}\rangle$, $\langle n^{(0)}|(H_0 - E_n^{(0)}) = 0$ which leads to

$$\langle n^{(0)}|\Delta_n(\lambda) - \lambda\hat{V}|n\rangle_\lambda = 0$$

Telling us that $|n^{(0)}\rangle$ is missing from the ket of the right-hand side. Now define the projection from

$$\mathbf{1} = \sum_k |k^{(0)}\rangle\langle k^{(0)}|$$

where the Hilbert space $\mathcal{H}$ has the basis $\{|k^{(0)}\rangle\}$ as

$$\phi_n = \mathbf{1} - |n^{(0)}\rangle\langle n^{(0)}| = \sum_{k \neq n} |k^{(0)}\rangle\langle k^{(0)}|$$

gives some n . Now,

$$\boxed{|n\rangle_\lambda = C_n(\lambda)|n^{(0)}\rangle + \frac{1}{E_n^{(0)} - \hat{H}_0} \phi_n(-\Delta_n(\lambda) + \lambda\hat{V})|n\rangle_\lambda}$$

Notice that $\langle n^{(0)}|n\rangle_\lambda = C_n(\lambda)$ and that $C_n(\lambda)$ dictates normalisation (how much $|n^{(0)}\rangle$ is in $|n\rangle_\lambda$). We expect the following

$$|n\rangle_\lambda = C_n(\lambda) \left(|n^{(0)}\rangle + \lambda|n^{(1)}\rangle + \lambda^2|n^{(2)}\rangle + \dots \right)$$

now we of course want to substitute.

LECTURE 4 (SEPTEMBER 10TH)

Recall. The perturbed ket satisfied the relation

$$(\lambda\hat{V} - \Delta_n(\lambda))|n\rangle_\lambda = (E_n^{(0)} - \hat{H}_0)|n\rangle_\lambda$$

Applying $\langle n^{(0)}|$,

$$\langle n^{(0)}|\lambda\hat{V}|n\rangle_\lambda = \Delta_n(\lambda)\langle n^{(0)}|n\rangle_\lambda \quad (\text{I})$$

we then obtain

$$|n\rangle_\lambda = C_n(\lambda)|n^{(0)}\rangle + \frac{1}{E_n^{(0)} - \hat{H}_0}\phi_n(\lambda\hat{V} - \Delta_n(\lambda))|n\rangle_\lambda \quad (\text{II})$$

and we have also obtained

$$C_n(\lambda) = \langle n^{(0)}|n\rangle_\lambda \quad (\text{III})$$

We now perform an expansion,

$$|n\rangle_\lambda = C_n(\lambda)\left(|n^{(0)}\rangle + \lambda|n^{(1)}\rangle + \lambda^2|n^{(2)}\rangle + \dots\right)$$

Applying equation (IV) to (I), we have

$$\langle n^{(0)}|\lambda\hat{V}|n^{(0)}\rangle + \lambda\langle n^{(1)}| + \lambda^2\langle n^{(2)}| + \dots = \Delta_n(\lambda) = \Delta_n^{(1)}\lambda + \Delta_n^{(2)}\lambda^2$$

Resultantly,

$$\boxed{\Delta_n^{(1)}\lambda = \langle n^{(0)}|\lambda\hat{V}|n^{(0)}\rangle}$$

Similary, wanting $\lambda^2\Delta_n^{(2)}$, we need to know $\lambda|n^{(1)}\rangle$. Equating (II) and (IV),

$$\lambda|n^{(1)}\rangle + \lambda^2|n^{(2)}\rangle + \dots = \frac{1}{E_n^{(0)} - H_0}\phi_n(\lambda\hat{V} - \Delta_n(\lambda))\left(|n^{(0)}\rangle + \lambda|n^{(1)}\rangle + \dots\right)$$

Leading us to

$$\boxed{\lambda|n^{(1)}\rangle = \frac{1}{E_n^{(0)} - H_0}\phi_n\lambda\hat{V}|n^{(0)}\rangle}$$

Applying to the above logic,

$$\langle n^{(0)}|\lambda\hat{V}|\lambda n^{(1)}\rangle = \Delta_n^{(2)}\lambda^2$$

and

$$\boxed{\Delta_n^{(2)}\lambda^2 = \langle n^{(0)}|\lambda\hat{V}\frac{1}{E_n^{(0)} - H_0}\phi_n\lambda\hat{V}|n^{(0)}\rangle}$$

Well, how can I express this?

$$\Delta_n^{(2)} \lambda^2 = \sum_{k \neq n} \frac{|\langle k^{(0)} | \lambda \hat{V} | n^{(0)} \rangle|^2}{E_n^{(0)} - E_k^{(0)}}$$

We cannot ignore $|n\rangle_\lambda$. We know that

$$\begin{aligned} {}_\lambda \langle n | n \rangle_\lambda = 1 &= |C_n(\lambda)|^2 \left(\langle n^{(0)} | + \lambda \langle n^{(1)} | + \lambda^2 \langle n^{(2)} | + \dots \right) \left(|n^{(0)}\rangle + \lambda |n^{(1)}\rangle + \lambda^2 |n^{(2)}\rangle + \dots \right) \\ &= |C_n(\lambda)|^2 \left(1 + \lambda^2 \langle n^{(1)} | n^{(1)} \rangle + \dots \right) \end{aligned}$$

and

$$|C_n(\lambda)|^2 = \frac{1}{1 + \lambda^2 \langle n^{(1)} | n^{(1)} \rangle + \dots} = 1 - \lambda^2 \langle n^{(1)} | n^{(1)} \rangle$$

Example. (Simple harmonic oscillator) Consider the solved harmonic oscillator with basis vectors $|n^{(0)}\rangle$ and $E^{(0)} = \hbar\omega(n + 1/2)$. Applying a uniform electric field, $E\hat{x}$,

$$\lambda \hat{V} = (-e)(-Ex) = eEx$$

For the first order, we have

$$\Delta_n^{(1)} = \langle n | \lambda \hat{V} | n \rangle = \langle n | eE\hat{x} | n \rangle = 0$$

For the second order, we have

$$\Delta_n^{(2)} = \sum_{k \neq n} \frac{|\langle k^{(0)} | eE\hat{x} | n^{(0)} \rangle|^2}{E_n - E_k} = -\frac{(eE)^2}{2m\omega^2} < 0$$

We know that this result, however, is exact. This is because of the form of the Hamiltonian,

$$H_0 + \lambda V = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2 + eEx = \frac{p^2}{2m} + \frac{m\omega^2}{2}(x - K)^2 - \frac{(eE)^2}{2m\omega^2}$$

where K is a constant. This is known as the Stark effect. Note that in the ground state,

$$\Delta_n^{(2)} = \sum_{k \neq n} \frac{|\langle k | \lambda \hat{V} | n \rangle|^2}{E_n - E_k} < 0$$

LECTURE 5 (SEPTEMBER 15TH)

Definition. (Degenerate perturbation theory) Notice that multiple states occupy the same energy level for the hydrogen atom with a Coulomb potential (e^2/r). Contrarily from the previous case where we saw one ket per energy level, we will try to see what happens when we deal with Hilbert spaces with multiple states per energy level. The state space with

identical energy levels can be expressed as (via projection)

$$P_0 = \sum_D |l_0\rangle\langle l_0|$$

where D is a degenerate subspace. The identity can thus be written as

$$\mathbf{1} = P_0 + P_1$$

Obviously,

$$\hat{H}_0 P_0 = \hat{H}_0 \sum_l |l_0\rangle\langle l_0| = E_D^{(0)} P_0$$

and

$$P_0 \hat{H}_0 = \sum_l |l_0\rangle\langle l_0| \hat{H}_0 = E_D^{(0)} P_0$$

and the two operators commute. We ultimately want to solve

$$\hat{H}|l\rangle = (\hat{H}_0 + \lambda \hat{V})|l\rangle = E|l\rangle$$

we state that it can be expressed as

$$|l\rangle = P_0|l\rangle + P_1|l\rangle$$

Applying the Hamiltonian, we expect

$$(\hat{H}_0 + \lambda \hat{V})(P_0|l\rangle + P_1|l\rangle) = E(P_0|l\rangle) + E(P_1|l\rangle)$$

Applying once more the projection operator P_0 ,

$$(E_D^{(0)} + P_0 \lambda \hat{V} P_0)P_0|l\rangle + P_0 \lambda \hat{V} P_1|l\rangle = E P_0|l\rangle$$

we can of course subtract the most-left term and arrive at

$$(P_0 \lambda \hat{V} P_0)P_0|l\rangle + P_0 \lambda \hat{V} P_1|l\rangle = (E - E_D^{(0)})P_0|l\rangle$$

where the energy difference on the right is called the energy spitting δE_0 . Ignoring the second term as we want only the 1st order, we have

$$(P_0 \lambda \hat{V} P_0)P_0|l\rangle = \delta E_0 P_0|l\rangle$$

We can also apply P_1 from the left to prove that indeed the second term is of second order and beyond. Having

$$\begin{aligned} P_1 \lambda \hat{V} P_0|l\rangle + (\hat{H}_0 + P_1 \lambda \hat{V} P_1)P_1|l\rangle &= E P_1|l\rangle \\ (E - \hat{H}_0 - P_1 \lambda \hat{V} P_1)P_1|l\rangle &= P_1 \lambda \hat{V} P_0|l\rangle \end{aligned}$$

which leads to

$$P_1|l\rangle = \frac{1}{E - H_0} P_1 \lambda V P_0 |l\rangle$$

Theorem. The following

$$\boxed{(P_0 \lambda V P_0) P_0 |l\rangle = \delta E_0 P_0 |l\rangle}$$

is ultimately the eigenvalue problem we want to solve to find the perturbed eigenkets. The energy is resultantly

$$\boxed{E = E_0^{(0)} + \delta E_0 + \lambda \sum_{k \notin D} \frac{|V_{kl}|^2}{E_D^{(0)} - E_k^{(0)}}$$

Example. (Isotropic rigid rotator) Consider the Hamiltonian

$$\hat{H}_0 = \frac{\vec{L} \cdot \vec{L}}{2I}$$

We know that the eigenkets and their energy levels are given as

$$|l, m\rangle \quad \& \quad \frac{l(l+1)\hbar^2}{2I}$$

If $l \neq 0$, degenerate $2l+1$. The perturbation will be given by

$$\hat{V} = -\vec{\mu} \cdot \vec{B} = -\frac{gqB}{2m} L_z$$

where $\vec{B}$ is an external magnetic field and $\vec{\mu} = gq\vec{L}/2m$. The g is dictated by whether its electron spin ($=2$) or whether its orbitals ($=1$). These kind of potentials are from Zeeman coupling. Now, we see that the potential is also commutative with L_z , as $[V, L_z] = 0$. We should utilise this! For a general state vector,

$$\langle m|[V, L_z]|m'\rangle = \langle m|VL_z - L_zV|m'\rangle = 0$$

which implies that

$$(m' - m)\hbar\langle m|V|m'\rangle = 0$$

Thus we have found the matrix element for the perturbation! We thus find

$$\frac{l(l+1)\hbar^2}{2I} - x\hbar m$$

for $x > 0$.

Definition. (Stark effect of the hydrogen atom) Assume no spin. For an Hydrogen atom in an electric field in the z -axis,

$$H_1 = (-e)V(r) = (-e)(-Ez) = eEz$$

LECTURE 6 (SEPTEMBER 17TH)

Proposition. The real hydrogen atom is only truly solved when dealing with the Dirac equation and the potential

$$\frac{-e^2}{r}$$

The spin-orbit interaction is given by

$$\begin{cases} \vec{E}' = \gamma(\vec{E} + \vec{\beta} \times \vec{B}) - \frac{\gamma^2}{\gamma + 1} \vec{\beta}(\vec{\beta} \cdot \vec{E}) \\ \vec{B}' = \gamma(\vec{B} - \vec{\beta} \times \vec{E}) \end{cases}$$

In the electron's rest frame, the lab frame variables $(\vec{E}, \vec{B})$ becomes $(\vec{E}', \vec{B}')$. Here,

$$\vec{\beta} = \vec{v}/c \quad \& \quad \gamma = \frac{1}{\sqrt{1 - \beta^2}}$$

Take $\vec{B} = 0$, we have

$$\begin{cases} \vec{E}' = \vec{E} \\ \vec{B}' = -\vec{\beta} \times \vec{E} = -\frac{\vec{v}}{c^2} \times \vec{E} = -\frac{\vec{v}}{c^2} \times (-\nabla\phi) = \frac{\vec{v}}{c^2} \times \hat{r} \frac{\partial\phi}{\partial r} \end{cases}$$

We have

$$H_{SO} = -\vec{\mu}_e \cdot \vec{B}'$$